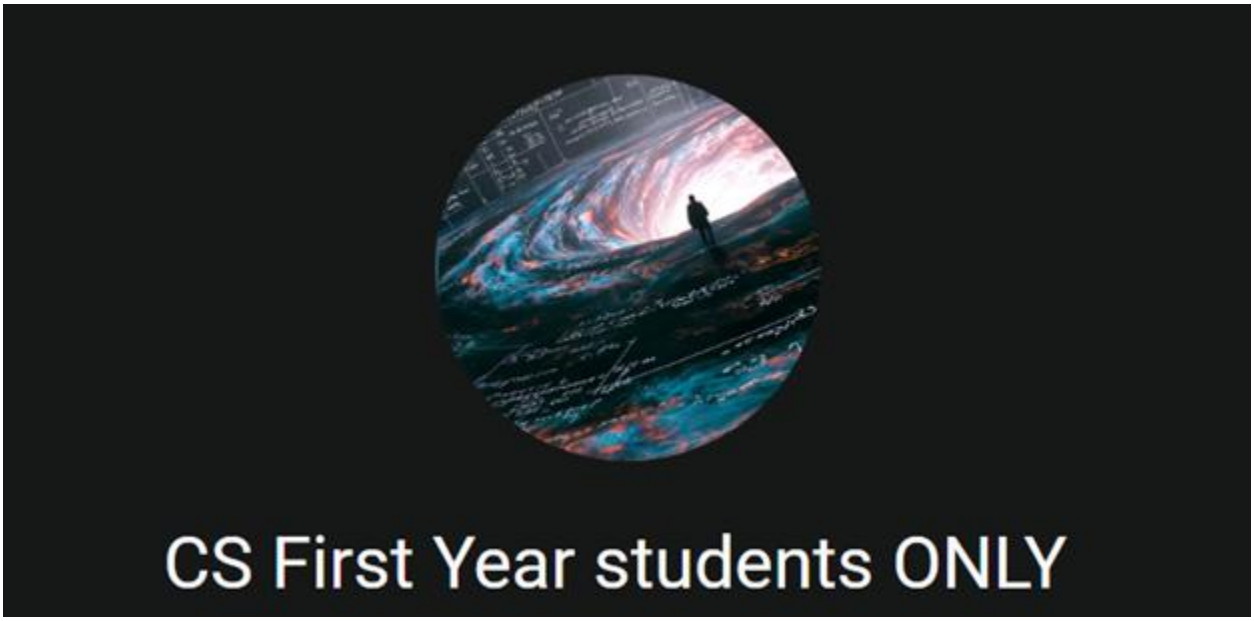
Midterm Exam For Second Semester (2024/2025)



Leader's :

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Ministry of High Education
Obour High Institute for Management & Informatics

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Midterm Exam for Second Semester (2024/2025)

Course Name: Discrete Mathematics	Course Code: BS103	Bylaw: 2024
Year: First	Program: Computer Sciences	
Date of Exam: Sunday 23/3/2025	Time Allowed: 1 hour	Total Points: 20
Course Instructor: Assist. Prof. Dr/ Saeed Hemeda	No. of questions: (2)	No. of pages: (1)

Exam instructions

- Make sure that the question form is identical to the answer sheet form
- The answer sheet must be in good condition from the moment received until it is delivered after taking the exam
- Do not bend the answer sheet or lean on it while answering

Answer the following questions:

Question One: [10 points] (ILOs: a1, a2, a3, b1, b2, c1, c2, d1)

Complete the following:

1) Suppose A and B are two non-empty sets then; $B + B^c B =$.....
2) Let's consider the interpretation v where; $v(p) = F, v(q) = T, v(r) = T$. Then the following propositional formula $[\sim(\sim p \rightarrow \sim q) \wedge r]$ is .....
3) If p is true and q is false, then $(\sim p \vee q) \rightarrow p$ is .....
4) If p, q, r are three switches then; $(p + q)(p + r)$=.....

Question Two: [10 points] (ILOs: a2, a4, b1, b3, b4, c1, c3, d1, d2)

1) Deduce the result of the compound statement
$[(K \rightarrow N) \wedge (N \rightarrow S)] \rightarrow (K \rightarrow S)$.
2) Consider the argument:

Premise: If you bought bread, then you went to the store
Premise: You bought bread
Conclusion: You went to the store

Analyze it using a truth table by representing each of the premises symbolically and show is the conclusion is valid or not.

Modus Ponens

End of Questions
Good Luck,,,
Examiner:
Assist. Prof. Dr/ Saeed Hemeda